

THERMAL REVISION DECK

PART 1: CORE THERMODYNAMICS & COOLING EQUATIONS

✂ **STEP 1:** Cut along outer dashed lines.

📄 **STEP 2:** Fold vertically along center blue dotted line.

🕒 **STEP 3:** Punch top-left ring hole.

✂ CUT BORDER

CARD #01

THERMODYNAMICS



NEWTON'S LAW OF COOLING

Question: What fundamental equation governs the rate of heat loss of a hot core capsule submerged in a cooler ambient fluid?

FOLO

FRONT: QUESTION

SPHERE STUDY DECK

CARD #01

KEY FORMULA



EXPONENTIAL DECAY

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

Explanation: The cooling rate $\frac{dT}{dt}$ is directly proportional to the instantaneous temperature gap $(T - T_{\text{env}})$.

BACK: REVEAL & LAW

k = cooling constant

✂ CUT BORDER

CARD #02

THERMODYNAMICS



DECAY CONSTANT (k)

Question: What does the constant k quantify physically, and which capsule is better insulated: $k = 0.15 \text{ min}^{-1}$ or $k = 0.015 \text{ min}^{-1}$?

FOLO

FRONT: QUESTION

SPHERE STUDY DECK

CARD #02

INSULATION FACTOR



CAPSULE B ($k = 0.015$)

Answer: k measures heat loss speed ($k = \frac{hA}{mc}$). A **smaller k value** means slower cooling and superior thermal insulation (heat held 10x longer!).

BACK: REVEAL & LAW

Units: min^{-1} or s^{-1}

✂ CUT BORDER

CARD #03

HEAT TRANSFER



FOURIER'S CONDUCTION LAW

Question: How does thermal energy transfer directly through solid matter, and what formula defines conduction heat flow?

FOLO

FRONT: QUESTION

SPHERE STUDY DECK

CARD #03

CONDUCTION EQUATION



MOLECULAR VIBRATION

$$q = -k_{\text{mat}} \cdot A \cdot \frac{dT}{dx}$$

Explanation: Kinetic energy spreads by vibrating atoms bumping adjacent particles. Low- k_{mat} materials (cotton, fiberglass) act as thermal insulators.

BACK: REVEAL & LAW

k_{mat} = conductivity

✂ CUT BORDER

CARD #04

EQUILIBRIUM



THERMAL EQUILIBRIUM

Question: When does net heat transfer between the capsule core and the 0°C ice bath drop to zero? What physical law governs this?

FOLO

FRONT: QUESTION

SPHERE STUDY DECK

CARD #04

ZEROTH LAW



EQUAL TEMPERATURES

Answer: When $T_{\text{capsule}} = T_{\text{env}} = 0^\circ\text{C}$. At this point, temperature gradient $\Delta T = 0$, so $\frac{dQ}{dt} = 0$. Governed by the **Zeroth Law of Thermodynamics**.

BACK: REVEAL & LAW

Net $\Delta Q = 0$ Joules



PART 2: RADIATIVE SHIELDING & MULTI-LAYER INSULATION (MLI)

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✂ CUT BORDER

CARD #05

RADIATION



SPACE VACUUM RADIATION

Question: Why is thermal radiation the **ONLY** heat transfer mode that can operate across the vacuum of deep space?

FOLD

FRONT: QUESTION

SPHERE STUDY DECK

CARD #05

STEFAN-BOLTZMANN



PHOTON HEAT FLOW

$$P_{\text{rad}} = \epsilon \sigma A (T^4 - T_{\text{env}}^4)$$

Answer: Vacuum contains zero molecules for conduction or convection. Heat radiates via infrared electromagnetic waves carrying energy without matter!

BACK: REVEAL & LAW

$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$

✂ CUT BORDER

CARD #06

SHIELDING



MYLAR RADIATIVE SHIELD

Question: How does a thin shiny aluminized Mylar foil protect spacecraft and astronauts from extreme solar thermal radiation?

FOLD

FRONT: QUESTION

SPHERE STUDY DECK

CARD #06

OPTICAL REFLECTION



INFRARED REFLECTANCE

Answer: Mylar's vapor-deposited aluminum coating has high reflectivity (> 90%) and very low emissivity ($\epsilon < 0.05$), bouncing infrared rays back like an optical mirror.

BACK: REVEAL & LAW

Reflects >90% IR heat

✂ CUT BORDER

CARD #07

CONVECTION



CONVECTIVE AIR TRAPPING

Question: Why is bubble wrap such an effective insulator during atmospheric lab trials? What mechanism does it eliminate?

FOLD

FRONT: QUESTION

SPHERE STUDY DECK

CARD #07

STAGNANT CELLS



STAGNANT AIR POCKETS

Answer: Air is a poor conductor ($k_{\text{air}} \approx 0.026 \text{ W/mK}$). Sealed bubble cells prevent **buoyancy circulation currents** from moving heat away!

BACK: REVEAL & LAW

Rayleigh convection stopped

✂ CUT BORDER

CARD #08

MLI SPACESUITS



MULTI-LAYER INSULATION

Question: Why do spacesuits and satellites stack Cotton + Bubble Wrap + Mylar in composite layers?

FOLD

FRONT: QUESTION

SPHERE STUDY DECK

CARD #08

TRIPLE SHIELD



TRIPLE HEAT SHIELD

Answer: Composite synergy:

- **Cotton:** Blocks solid conduction.
- **Bubble Wrap:** Traps fluid convection.
- **Mylar:** Reflects infrared radiation.

BACK: REVEAL & LAW

All 3 heat paths blocked

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✂ CUT BORDER

CARD #09

SIMULATION

 **DIGITAL TWIN CONCEPT**

Question: What is a Digital Twin in modern aerospace engineering, and how is it used in SPHERE Mission Control?

FRONT: QUESTION SPHERE STUDY DECK

CARD #09

VIRTUAL TELEMETRY

 **LIVE VIRTUAL MODEL**

Answer: A high-fidelity physics model running equations in real-time alongside live physical sensors, enabling immediate anomaly detection and residual tracking.

BACK: REVEAL & LAW Hardware + Simulation

✂ CUT BORDER

CARD #10

LAB ANALOG

 **ICE BATH SPACE ANALOG**

Question: Why do we submerge the insulated capsule into a 0°C ice-water slurry during laboratory mission runs?

FRONT: QUESTION SPHERE STUDY DECK

CARD #10

CONSTANT SINK

 **CONSTANT HEAT SINK**

Answer: Melting ice water stays at an exact, invariant 0°C ($T_{\text{env}} = 0^\circ\text{C}$), simulating the immense, infinite thermal heat sink of deep space.

BACK: REVEAL & LAW $T_{\text{env}} = 0.0^\circ\text{C}$ constant

✂ CUT BORDER

CARD #11

THERMAL MASS

 **HEAT CAPACITY OF WATER**

Question: Why is liquid water ($c = 4.184 \text{ J/g}^\circ\text{C}$) ideal for capsule core telemetry trials? What is the heat formula?

FRONT: QUESTION SPHERE STUDY DECK

CARD #11

THERMAL ENERGY

 **HIGH THERMAL MASS**

$$Q = m \cdot c \cdot \Delta T$$

Answer: Water stores large thermal energy per gram, producing smooth, measurable exponential decay over 15+ minutes without dropping instantly.

BACK: REVEAL & LAW $c = 4184 \text{ J/kg}\cdot\text{K}$

✂ CUT BORDER

CARD #12

GRAPH ANALYSIS

 **DECAY CURVE CURVATURE**

Question: Why is the temperature-time graph $T(t)$ steep at $t = 0$ and increasingly flat toward $t = 15 \text{ min}$?

FRONT: QUESTION SPHERE STUDY DECK

CARD #12

ASYMPTOTIC FLOW

 **SHRINKING TEMP GAP**

Answer: Heat flux rate $\frac{dQ}{dt} \propto (T - T_{\text{env}})$. As the capsule cools, $(T - T_{\text{env}})$ shrinks, causing the rate of temperature drop to slow down asymptotically.

BACK: REVEAL & LAW Slope = $-k(T - T_{\text{env}})$

THERMAL REVISION DECK

PART 4: MISSION PROTOCOLS, MATH METRICS & SYNTHESIS

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CUT BORDER

CARD #13

METRICS

CORRELATION SCORE (R^2)

Question: How does Mission Control calculate the Correlation Score between live physical sensor telemetry and the Digital Twin model?

FRONT: QUESTION

SPHERE STUDY DECK

CARD #13

MODEL ACCURACY

RESIDUAL VARIANCE

$$R^2 = 1 - \frac{\sum (T_{\text{meas}} - T_{\text{pred}})^2}{\sum (T_{\text{meas}} - \bar{T})^2}$$

Answer: Quantifies goodness-of-fit. A score > 95% proves the theoretical physics model matches actual mission hardware telemetry with high fidelity.

BACK: REVEAL & LAW

Target: >95% Correlation

CUT BORDER

CARD #14

LAB PROTOCOL

PROBE PLACEMENT

Question: Why must the digital thermocouple probe be centered inside the fluid rather than touching the capsule wall?

FRONT: QUESTION

SPHERE STUDY DECK

CARD #14

BOUNDARY LAYER

BULK FLUID TEMPERATURE

Answer: Touching the outer wall measures the cold boundary gradient rather than core liquid bulk temperature, introducing measurement artifacts into telemetry.

BACK: REVEAL & LAW

Center probe exactly

CUT BORDER

CARD #15

SI UNITS

SI UNITS REFERENCE

Question: What are the standard SI units for temperature T , heat flow Q , thermal conductivity k_{mat} , and cooling constant k ?

FRONT: QUESTION

SPHERE STUDY DECK

CARD #15

STANDARD UNITS

UNITS CHEAT SHEET

- Temperature (T): Kelvin (K) or Celsius ($^{\circ}\text{C}$)
- Thermal Energy (Q): Joules (J)
- Decay Constant (k): s^{-1} or min^{-1}
- Conductivity (k_{mat}): $\text{W}/(\text{m} \cdot \text{K})$

BACK: REVEAL & LAW

Standard Metric Units

CUT BORDER

CARD #16

CREW STRATEGY

CREW MLI OPTIMIZATION

Question: Which insulation configuration produces the highest survival time and smallest decay constant k ? Why?

FRONT: QUESTION

SPHERE STUDY DECK

CARD #16

SURVIVAL RECORD

3-LAYER COMPOSITE MLI

Answer: Cotton + Bubble Wrap + Mylar drops decay constant k from $\sim 0.150 \text{ min}^{-1}$ (bare tube) down to $< 0.020 \text{ min}^{-1}$, preserving capsule heat for > 20 minutes.

BACK: REVEAL & LAW

Maximum Thermal Shield